Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 06

Subject: BSC102/AP | **Date: 09/11/2026**
Max Marks: 10 | **Deadline: 20/11/2026**

CO6: Learner will be able to CLASSIFY semiconductors and EXPLAIN variation of Fermi level with temperature and doping concentration.

Problem 1

Calculate the probabilities for an electronic state to be occupied at 20°C, if the energy of these states lies 0.11 eV above and 0.11 eV below the Fermi level.

Ans. ($f(E)$ for above $= 0.0126$, $f(E)$ for below $= 0.9874$)

Solution:

The Fermi-Dirac distribution function gives the probability of occupation:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

Given:

- $T = 20°\text{C} = 20 + 273 = 293$ K
- Boltzmann constant: $k_B = 8.617 \times 10^{-5}$ eV/K
- Thermal energy:

$$k_B T = (8.617 \times 10^{-5}\ \text{eV/K}) \times 293\ \text{K} \approx 0.02525\ \text{eV}$$

(i) For the state lying 0.11 eV above the Fermi level ($E - E_F = +0.11$ eV):

$$\frac{E - E_F}{k_B T} = \frac{0.11\ \text{eV}}{0.02525\ \text{eV}} \approx 4.3564$$

$$f(E) = \frac{1}{1 + e^{4.3564}} = \frac{1}{1 + 77.976} = \frac{1}{78.976} \approx \mathbf{0.01266} \approx \mathbf{0.0126}$$

(ii) For the state lying 0.11 eV below the Fermi level ($E - E_F = -0.11$ eV):

$$\frac{E - E_F}{k_B T} = -\frac{0.11 \text{ eV}}{0.02525 \text{ eV}} \approx -4.3564$$

$$f(E) = \frac{1}{1 + e^{-4.3564}} = 1 - f(E_{\text{above}}) = 1 - 0.0126 = \mathbf{0.9874}$$

Problem 2

In a solid, consider the energy level lying 0.01 eV below Fermi level. What is the probability of this level not being occupied by an electron?

Solution:

The probability that an energy state is not occupied by an electron is given by:

$$P(\text{not occupied}) = 1 - f(E) = 1 - \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} = \frac{1}{1 + \exp\left(\frac{E_F - E}{k_B T}\right)}$$

Given:

- Energy difference: $E_F - E = 0.01$ eV
- Standard room temperature: $T = 300$ K
- Thermal energy: $k_B T = 8.617 \times 10^{-5} \times 300 \approx 0.02585$ eV (or using $k_B = 1.38 \times 10^{-23}$ J/K, $k_B T = 0.02586$ eV)

Evaluating the argument:

$$\frac{E_F - E}{k_B T} = \frac{0.01 \text{ eV}}{0.02525 \text{ eV}} \approx 0.3957 \quad \text{(at 293 K)}$$

Using room temperature ($T = 300$ K, where $k_B T = 0.02585$ eV):

$$\frac{E_F - E}{k_B T} = \frac{0.01}{0.02585} \approx 0.3868$$

$$1 - f(E) = \frac{1}{1 + e^{0.3957}} = \frac{1}{1 + 1.4854} = \frac{1}{2.4854} \approx 0.4023$$

Thus, the probability of the state not being occupied is **0.4023**.

Problem 3

The intrinsic carrier density is 1.5×10^{16} m^{-3}. If the mobility of the electron and hole are 0.13 and 0.05 m^2 V^{-1} s^{-1}, calculate the conductivity.

Solution:

The electrical conductivity of an intrinsic semiconductor is:

$$\sigma = n_i e(\mu_e + \mu_h)$$

Given data:

- Intrinsic carrier concentration: $n_i = 1.5 \times 10^{16}\ \mathrm{m}^{-3}$
- Electron charge: $e = 1.6 \times 10^{-19}\ \mathrm{C}$
- Electron mobility: $\mu_e = 0.13\ \mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$
- Hole mobility: $\mu_h = 0.05\ \mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$

Substituting the values:

$$\begin{aligned}\sigma &= (1.5 \times 10^{16}\ \mathrm{m}^{-3}) \times (1.6 \times 10^{-19}\ \mathrm{C}) \times (0.13 + 0.05)\ \mathrm{m}^2/(\mathrm{V}\cdot\mathrm{s}) \\ &= (2.4 \times 10^{-3}) \times (0.18) \\ &= \mathbf{4.32 \times 10^{-4}}\ \mathrm{S/m} \quad (\text{or } \Omega^{-1}\cdot\mathrm{m}^{-1})\end{aligned}$$

Note: The printed answer contains a typographical error in the exponent (10^4 instead of 10^{-4}); the correct evaluated value is 4.32×10^{-4} S/m.

Problem 4

The Intrinsic carrier density at room temperature in Ge is $2.37 \times 10^{19}\ \mathrm{m}^{-3}$ if the electron and hole mobilities are 0.38 and 0.18 $\mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ respectively, calculate the resistivity.

Solution:

The intrinsic conductivity is:

$$\sigma = n_i e(\mu_e + \mu_h)$$

Given:

- $n_i = 2.37 \times 10^{19}\ \mathrm{m}^{-3}$
- $e = 1.6 \times 10^{-19}\ \mathrm{C}$
- $\mu_e = 0.38\ \mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$
- $\mu_h = 0.18\ \mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$

Calculating conductivity:

$$\begin{aligned}\sigma &= (2.37 \times 10^{19}) \times (1.6 \times 10^{-19}) \times (0.38 + 0.18) \\ &= (3.792) \times (0.56) \\ &= 2.12352\ \mathrm{S/m}\end{aligned}$$

The resistivity ρ is the reciprocal of conductivity:

$$\rho = \frac{1}{\sigma} = \frac{1}{2.12352 \text{ S/m}} \approx \mathbf{0.4709\,\Omega \cdot m} \approx \mathbf{0.471\,\Omega \cdot m}$$

Problem 5

In a P-type germanium, $n_i = 2.1 \times 10^{19}$ m^{-3}, density of boron 4.5×10^{23} atoms/m^3. The electron and hole mobility are 0.4 and 0.2 m^2 V^{-1} s^{-1} respectively. What is its conductivity before and after addition of boron atoms?

Solution:

(i) Conductivity before addition of boron (intrinsic):

$$\sigma_{\text{before}} = n_i e(\mu_e + \mu_h)$$

Given:

- $n_i = 2.1 \times 10^{19}$ m^{-3}
- $e = 1.6 \times 10^{-19}$ C
- $\mu_e = 0.4$ m^2 V^{-1} s^{-1}
- $\mu_h = 0.2$ m^2 V^{-1} s^{-1}

$$\sigma_{\text{before}} = (2.1 \times 10^{19}) \times (1.6 \times 10^{-19}) \times (0.4 + 0.2) = 3.36 \times 0.6 = \mathbf{2.016} \text{ S/m}$$

(ii) Conductivity after addition of boron (extrinsic p-type): Boron is an acceptor impurity ($N_A = 4.5 \times 10^{23}$ m^{-3}). Assuming complete ionization, the majority hole concentration is:

$$p \approx N_A = 4.5 \times 10^{23} \text{ m}^{-3}$$

Since $N_A \gg n_i$, conduction is dominated by holes:

$$\begin{aligned}\sigma_{\text{after}} &\approx p \cdot e \cdot \mu_h \\ &= (4.5 \times 10^{23} \text{ m}^{-3}) \times (1.6 \times 10^{-19} \text{ C}) \times (0.2 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}) \\ &= (7.2 \times 10^4) \times 0.2 \\ &= \mathbf{1.44 \times 10^4} \text{ S/m}\end{aligned}$$